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Realizing High Capacity and Zero Strain in Layered Oxide Cathodes via Lithium Dual-Site Substitution for Sodium-Ion Batteries

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ABSTRACT: Sodium-ion batteries have garnered unprecedented attention as an electrochemical energy storage technology, but it remains challenging to design high-energy-density cathode materials with low structural strain during the dynamic (de)sodiation processes. Herein, we report a P2-layered lithium dual-site substituted $Na_{0.7}Li_{0.03}[Mg_{0.15}Li_{0.07}Mn_{0.75}]O_2$ (NMLMO) cathode material, in which Li ions occupy both transition-metal (TM) and alkali-metal (AM) sites. The combination of theoretical calculations and experimental characterizations reveals that Li_{TM} creates Na–O–Li electronic configurations to boost the capacity derived from the oxygen anionic redox, whilst Li_{AM} serves as LiO_6 prismatic pillars to stabilize the layered structure through suppressing the detrimental phase transitions. As a result, NMLMO delivers a high specific capacity of 266 mAh g^{-1} and simultaneously exhibits the nearly zero-strain characteristic within a wide voltage range of 1.5–4.6 V. Our findings highlight the effective way of dual-site substitution to break the capacity-stability trade-off in cathode materials for advanced rechargeable batteries.

INTRODUCTION

Propelled by the high-density storage and the efficient utilization of renewable energy, sodium-ion batteries (SIBs) have emerged as promising alternatives to their lithium-ion counterparts because of the cost advantage and sustainability attributes.[1–5] The SIB technology urgently calls for high-capacity and long-life cathode materials to meet the large-scale application, especially in electrochemical energy storage and conversion systems.[6–8] Various cathode materials have been extensively investigated, including transition metal oxides, polyanionic frameworks, Prussian blue analogs and organic compounds.[9–16] Among these candidates, layered sodium transition metal oxides (Na_xTMO_2, $0 < x \leq 1$) are particularly attractive owing to their unique intercalation mechanism, high theoretical capacity and flexible composition.[17] However, the amount of Na ions that can reversibly (de)intercalate in/out of the oxide hosts is strictly limited due to the trade-off between specific capacity and structural stability, which poses a formidable challenge: the simultaneous optimization of energy density and electrochemical stability for SIB cathode materials.

In terms of energy density, both high capacity and high voltage can be achieved in Na_xTMO_2 cathodes through the substitution of transition-metal (TM) cations with A (A: alkali or alkali-earth) ions such as Li^+ and Mg^{2+}, where the oxygen anionic redox (OAR) reaction is triggered during Na^+ extraction and insertion.[18–20] The OAR activity arises from the formation of local Na–O–A configurations, which can activate the non-bonding O 2*p* orbitals (states) below the Fermi level to participate in the charge compensation, thus realizing the extra capacity beyond the traditional TM redox.[21] Even though the OAR chemistry establishes a new redox paradigm to access high energy density for $Na_x[A_yTM_{1-y}]O_2$ ($0 < y < 1$) cathodes, it comes at a price of structural instability, particularly in high state-of-charge regions.[22] Upon high voltage (> 4.0 V *vs.* Na^+/Na), the P-type (P: Prismatic NaO_6 unit) structures generally undergo the P-O (O: Octahedral NaO_6 unit) or P-OP (OP: O/P intergrowth) phase transition, which is engendered by the TMO_2 slabs gliding on account of the enhanced electrostatic repulsion between the eclipsed oxygen ions during Na^+ deintercalation.[23,24] Under low voltage (< 2.0 V *vs.* Na^+/Na), the cooperative Jahn-Teller effect of high spin Mn^{3+} (t_{2g}^3-e_g^1) in P2-type Mn-based compounds drives a geometrical distortion of the octahedral MnO_6 units, resulting in a phase transformation from the hexagonal polymorph to an orthorhombic phase (P'2) during Na^+ intercalation.[25,26] Moreover, OAR-active cathodes are also associated with a series of local disorders, such as O–O dimer formation and lattice oxygen release, which renders the OAR process partially reversible.[27–30] These structural evolutions mentioned above will inevitably bring about anisotropic lattice strain and dramatic volume change, further deteriorating the battery performance during electrochemical cycling.[31,32]

With regards to the improvement of electrochemical stability for Na-layered oxide cathodes, a straightforward way is to limit the cut-off voltage range within 2.0–4.0 V, as more electrostatic shielding Na ions can be maintained in the alkali-metal (AM) layers to stabilize the layered structure.[33] Nevertheless, narrowing the operating voltage window tends to evade the occurrence of oxygen redox, compromising the available voltage and specific capacity. In addition, it is well recognized that constructing a zero-strain host that exhibits extremely small lattice volume variation during the dynamic (de)intercalation processes, is considered as the optimal route to ameliorate the structural and electrochemical stability because of the minimal disruption to Na^+ transport pathways.[34] Although introducing pillar ions (*e.g.*, K^+, Mg^{2+}, Zn^{2+}, Ca^{2+} and Fe^{3+}) into the AM layers can effectively mitigate the unfavorable phase transitions and reduce the volume variation through the pinning effect, it still remains challenging for OAR-active cathodes to simultaneously realize high capacity and zero strain.[35–39] Therefore, overcoming the fundamental capacity-stability paradox is paramount for the practical implementation of cathode materials with cationic and anionic redox.

Here, we propose a P2-layered Li dual-site substituted $Na_{0.7}Li_{0.03}[Mg_{0.15}Li_{0.07}Mn_{0.75}]O_2$ (NMLMO) cathode material with the high-capacity and zero-strain characteristic. The two explicit roles of Li_{TM} (Li ions in the octahedral TM sites) and Li_{AM} (Li ions in the trigonal prismatic AM sites) have been unraveled by using density functional theory calculations, advanced analytical techniques and in-situ characterizations. The data show that Li_{TM} forms local Na–O–Li configurations to harvest the OAR-boosted capacity, while Li_{AM} acts as rigid LiO_6 pillars to eliminate the P2-OP4 and P2-P'2 phase transitions during cycling. Benefitting from the synergistic effect of Li_{TM} and Li_{AM}, NMLMO not only delivers a high specific capacity of 266 mAh g^{-1}, but also presents a complete single-phase solid-solution reaction with only 1.2% lattice volume variation in a wide voltage range of 1.5–4.6 V. The dual-site substitution strategy paves a way for the rational design of high-energy-density cathode materials.

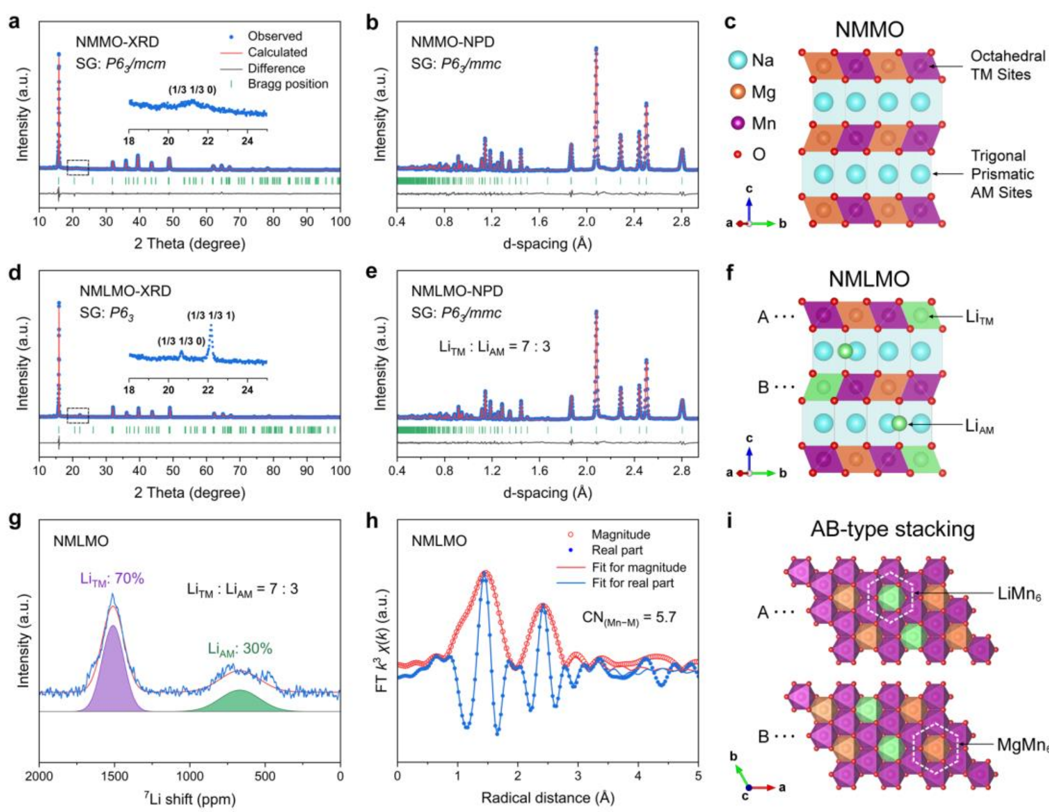

Figure 1. Crystal structure of NMMO and NMLMO. Rietveld refined XRD patterns of (a) NMMO and (d) NMLMO. Insets of a and d highlight the superlattice peak region. Rietveld refined NPD patterns of (b) NMMO and (e) NMLMO. Schematic

illustration of (c) NMMO and (f) NMLMO projected in b-c plane. (g) Fitted ^{7}Li MAS NMR spectra of NMLMO. (h) Fitted Mn K-edge FT-EXAFS spectra of NMLMO. i, Schematic illustration of NMLMO projected in a-b plane.

RESULTS

Structural Characterization. $Na_{0.7}Mg_{0.15}Mn_{0.85}O_2$ (NMMO) and Li-substituted $Na_{0.7}Mg_{0.15}Li_{0.1}Mn_{0.75}O_2$ (NMLMO) were prepared through a solid-state reaction, as described in the Experimental Section (Supporting Information). Inductive coupled plasma optical emission spectroscopy (ICP-OES) analysis demonstrates that the chemical composition of the as-prepared samples is consistent with the expected stoichiometry (Table S1). Both NMMO and NMLMO adopt a hexagonal P2-type layered structure with in-plane honeycomb superlattice in the TM layers, as confirmed by X-ray diffraction (XRD) and neutron powder diffraction (NPD) (Figures 1a,b,d,e, S1 and S2). However, the difference in superlattice peaks indicates the stacking sequence of the honeycomb TM layers in NMLMO (AB-type) varies from that in NMMO (AA-type) (Figures 1c,f,i and S3).[40] The XRD patterns of both materials can be Rietveld refined using two structure models: one in a space group (SG) of $P6_3/mmc$, and the other in a SG of $P6_3/mcm$ (NMMO) or $P6_3$ (NMLMO) based on their diversity in the superlattice structure. The XRD and NPD Rietveld refinement results (Tables S2–S5) show that Mg and Mn ions occupy the octahedral TM sites for both materials, while 70% Li ions sit in the TM sites (Li_{TM}) and 30% Li ions reside in the AM sites (Li_{AM}) for NMLMO. The existence of Li_{TM} leads to a contraction of the c-lattice parameter because of the extra electrostatic attraction to the adjacent oxygen layers (Tables S6 and S7). Note that for the P2-type structures, Na ions occupy two groups of trigonal prismatic AM sites: Na_e and Na_f sites. Na_e sites share edges with TMO_6 octahedra and are electrostatically more stable than Na_f sites that share faces with TMO_6 octahedra in the two adjacent TMO_2 slabs.[41] Accordingly, Li_{AM} is assigned to the Na_e sites, which is supported by the Rietveld refined NPD results.

The two local environments of Li ions in NMLMO were further evidenced by ^{7}Li solid-state magic angle spinning (MAS) nuclear magnetic resonance (NMR) spectroscopy. As presented in Figure 1g, two resonance peaks at around 1500 and 700 ppm are ascribed to the Li ions sitting around six Mn ions ($LiMn_6$) in the TM layers and the ones residing in the AM layers, respectively.[42–44] The quantitative data illustrate that the content ratio of Li_{TM}:Li_{AM} is 7:3, matching well with the NPD results. To emphasize the Li distribution, the molecular formula of NMLMO is hereafter termed as $Na_{0.7}Li_{0.03}[Mg_{0.15}Li_{0.07}Mn_{0.75}]O_2$, where the Li elements in and out of the square brackets are located in the TM and AM layers, respectively. Besides, $Na_{0.7}Mg_{0.15}Li_xMn_{0.85-x}O_2$ (x = 0, 0.05, 0.07, 0.1. 0.13) with various Li substitution contents were prepared to identify the formation process of Li_{TM} and Li_{AM}. The XRD results (Figures S4–S6) reveal that Li ions firstly enter into the TM layers to form solid solution under low substitution content ($x < 0.05$). While, when the Li content is higher than 0.05 per formula unit, the additional Li ions gradually transfer into the AM layers, hence generating the Li dual-site substituted structures (Figure S7).

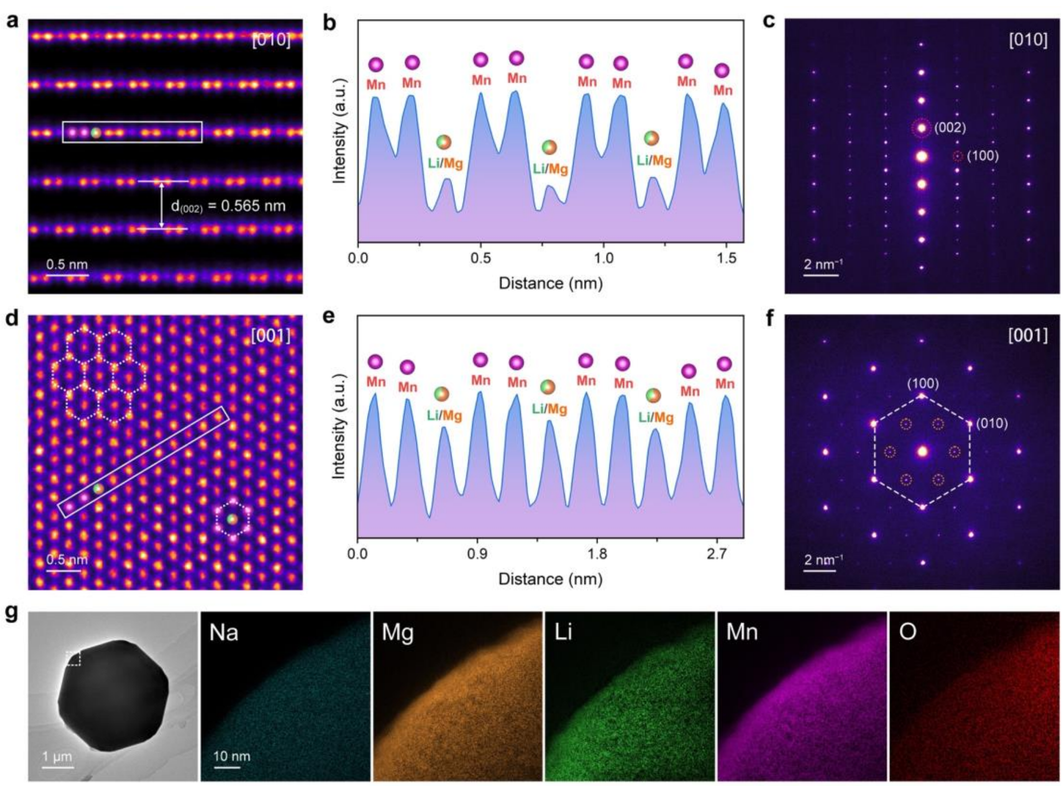

Figure 2. Honeycomb-layered superlattice and morphological characterization of NMLMO. HAADF-STEM images along the (a) [010] and (d) [001] zone axis. Intensity line profiles obtained from the area marked by white rectangles in (b) a and (e) d. SAED patterns along the (c) [010] and (f) [001] zone axis. The ordered superlattice is identified by orange dotted circles in f. (g) TEM image and the corresponding EDS elemental mappings in the marked region.

The oxidation state and local structural environment of Mn in NMMO and NMLMO were investigated by synchrotron-based X-ray absorption spectroscopy (XAS). Figure S8 exhibits the normalized Mn K-edge X-ray absorption near edge structure (XANES) spectra of NMMO and NMLMO. Using the linear relationship between the edge energy and the oxidation state, the average valences of Mn in NMMO and NMLMO are estimated to be 3.6+ and 3.9+, respectively, in line with the X-ray photoelectron spectroscopy (XPS) results (Figure S9). According to the fitting of Mn K-edge Fourier transformed (FT) extended X-ray absorption fine structure (EXAFS) spectra for NMLMO (Figure 1h), the coordination number (CN) of Mn–M (M = Mn, Mg or Li) interaction in the second coordination shell is calculated to be 5.7, which is lower than that of NMMO (6.0, the standard hexagonal coordination value) (Figure S10). This suggests that some TM vacancies are formed in the TM layers as a number of Li ions enter into the AM sites.

Both NMMO and NMLMO exhibit a hexagonal plate-type morphology with a particle size range of 2–4 μm (Figures S11–S13). The precise atomic-level structure of NMLMO was captured by high-angle annular dark-field (HAADF) spherical aberration-corrected scanning transmission electron microscopy (AC-STEM). The column positions of Mn (strong spots) and Li/Mg (weak spots) present in a typical honeycomb ordering from the perspective of the [010] (Figures 2a and S14) and [001] (Figures 2d and S15) zone axis. The honeycomb TM layers of NMLMO are staked in an offset arrangement along the c-axis, validating the AB-type superlattice stacking (Figure 2a). Meanwhile, the interlayer lattice distance between the adjacent TM layers is measured to be 0.565 nm, corresponding to the d-spacing value of the (002) planes for the P2 layered structure. The intensity line profiles (Figure 2b,e) extracted from the STEM images further identify the in-plane honeycomb superlattice in NMLMO. Moreover, the selected area electron diffraction (SAED) patterns (Figure 2c,f) along the [010] and [001] zone axis reconfirm the phase purity, hexagonal lattice symmetry and ordered superlattice structure. The energy dispersive spectroscopy (EDS) elemental mappings (Figure 2g) indicate Na, Mg, Li, Mn and O elements are uniformly distributed throughout the NMLMO particle.

Electrochemical Performance. The electrochemical properties of NMMO and NMLMO were evaluated versus Na^+/Na within the voltage window of 1.5–4.6 V. Figure 3a presents the first galvanostatic charge-discharge voltage profiles of NMMO and NMLMO at the rate of 0.05 C (10 mA g^{-1}). On charge, the voltage curve of NMMO shows a prominent slope region, followed by a high-voltage plateau at ~4.42 V. By comparison, the curve of NMLMO rapidly reaches a single long plateau at ~4.30 V, which contributes to almost the whole charge capacity of 197 mAh g^{-1}, corresponding to the removal of ~0.69 Na^+ per formula unit. It can be deduced that more oxygen ions participate in the charge compensation for NMLMO, as the high-voltage plateau generally stems from the oxygen oxidation process in oxygen-redox materials.[20] Moreover, a voltage drop of 0.12 V is found between the two high-voltage plateaus, suggesting the lower voltage of oxygen oxidation in NMLMO (Figures 3a and S16). Upon further discharging, NMLMO delivers a specific capacity of 266 mAh g^{-1} (equivalent to the reinsertion of 0.93 Na^+ per formula unit), which is much higher than that of NMMO (212 mAh g^{-1}). During the repeated cycles, the voltage profiles of NMMO show continuous change, especially in the high-voltage plateau region (Figure 3b,d). For NMLMO, the profiles are well overlapped in the early cycles and the oxygen oxidation-related plateau is preserved even after 30 cycles (Figure 3c,e). In addition, NMLMO exhibits an improved capacity retention of 80.9% compared to NMMO (57.1%) after 50 cycles at the rate of 0.5 C (Figure 3f), as well as a competitive rate performance (126 mAh g^{-1} at 5 C) among the reported P2-type oxide cathodes (Figure S17 and Table S8). The high specific capacity and superior cycling performance of NMLMO demonstrate that the highly active and reversible oxygen redox reaction is achieved by the Li dual-site substitution.

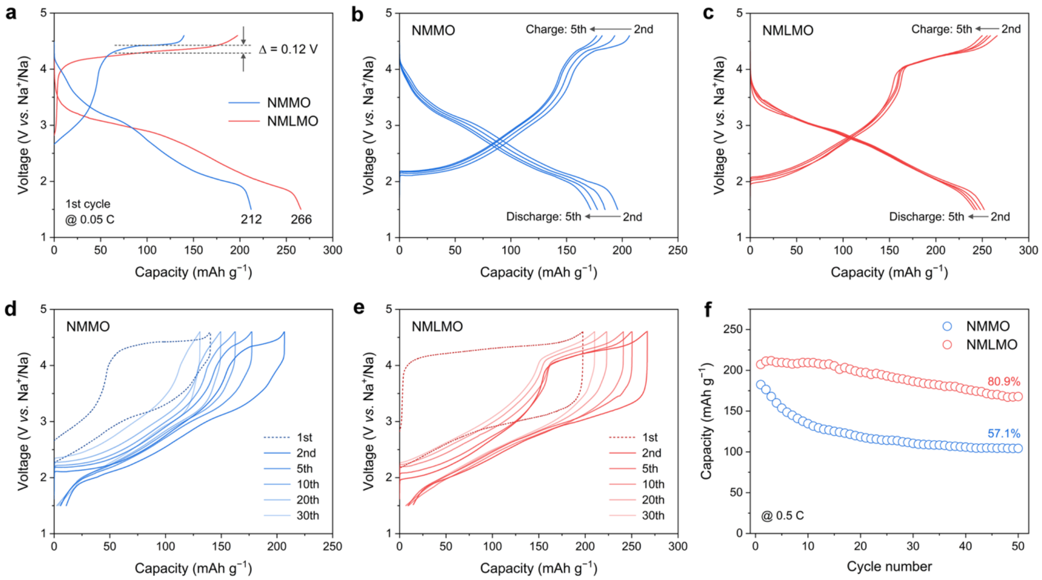

Figure 3. Electrochemical performance of NMMO and NMLMO in the voltage range of 1.5–4.6 V. (a) First galvanostatic charge-discharge voltage profiles at 0.05 C. Voltage profiles of (b) NMMO and (c) NMLMO in the 2–5 cycles at 0.05 C. Representative voltage profiles of (d) NMMO and (e) NMLMO during cycling at 0.05 C. (f) Cycling performance at 0.5 C.

Redox Mechanism. To explore the redox mechanism of NMMO and NMLMO, their electronic structures were firstly studied by the first principle density functional theory (DFT) calculations. The effect of Li_{TM} on the charge compensation is mainly taken into consideration for NMLMO, since the presence of A (alkali or alkali-earth) ions in the TM sites of Na-layered oxides can activate the oxygen redox reaction due to the formation of oxygen lone pairs associated with non-bonding O 2*p* states in the electronic structure.[45,46] Figure 4a,b shows the projected partial density of states (pDOS) and the corresponding schematic band diagrams for NMMO and NMLMO. Compared with the Na–O–Mg or Na–O–Mn interaction, more ionic Na–O–Li interaction places the O 2*p* states to an elevated position (closer to the Fermi level) in energy, which accounts for the occurrence of the oxygen redox at a relatively low charge voltage and the observed decrement of the oxygen-related high-voltage plateau for NMLMO. Additionally, these elevated O 2*p* orbitals are highly active and can donate or receive extra charges during Na^+ (de)intercalation. As a consequence, the introduction of Li ions in the TM sites not only facilitates the oxygen redox process, but also enhances the specific capacity of NMLMO.

Synchrotron-based XAS analysis was then performed to elucidate the evolution of the valance state of Mn in NMMO and NMLMO during the charge and discharge processes (Figure 4c–h). With charging from the open-circuit voltage (OCV) to 4.3 V, the Mn K-edge of both cathodes shifts toward higher energy, suggesting the oxidation of Mn^{3+}. According to the linear calibration of the XANES spectra, it is quantitatively estimated that the valences of Mn in NMMO and NMLMO vary from 3.6+ to 3.8+ and from 3.9+ to 4.0+, respectively. When the voltage further increases from 4.3 to 4.6 V, the Mn K-edge of NMMO moves to that of the reference MnO_2 (Mn^{4+}). While, the Mn K-edge of NMLMO shows no edge shift, which indicates no charge compensation of Mn occurs. In the following discharge process from 4.6 to 2.7 V, the Mn K-edge of NMMO shifts back to lower energy, demonstrating the reduction of Mn^{4+}. Again, the position of Mn-edge remains unchanged for NMLMO, which implies that the charge compensation is dominated by O rather than Mn. Upon further discharging to 1.5 V, the Mn K-edge of both cathodes illustrates a shift to lower energy and the estimated valences of Mn are 3.5+ and 3.7+ for NMMO and NMLMO, respectively. During recharging, the oxidation states of Mn undergo almost the same variation as those in the first charge process.

The corresponding charge compensation of Mn was also evidenced by Ar etching assisted XPS (Figure S18). To further ascertain the oxygen redox, O 1s XPS analysis was employed (Figure S19). The peaks at higher binding energies (~531.0 and ~532.0 eV) and lower binding energy (~529.2 eV) can be assigned to the surface oxygen-deposited species and the lattice oxygen ions (O^{2-}), respectively. For NMMO at the fully charged state (4.6 V), a new peak located at ~530.5 eV appears, which can be ascribed to the component of peroxo-like species ($(O_2)^{n-}$) formed by the oxidation of O^{2-}.[47] Comparatively, the $(O_2)^{n-}$ peak is introduced at 4.3 V for NMLMO, indicating the lower voltage of oxygen oxidation. Moreover, the ratio of $(O_2)^{n-}$ peak area over the fitted O 1s spectra of NMLMO at 4.6 V is much higher than that of NMMO, which suggests that much more oxygen ions participate in the charge compensation. Therefore, NMLMO presents high oxygen redox reactivity through the Li substitution, in accord with the DFT calculations.

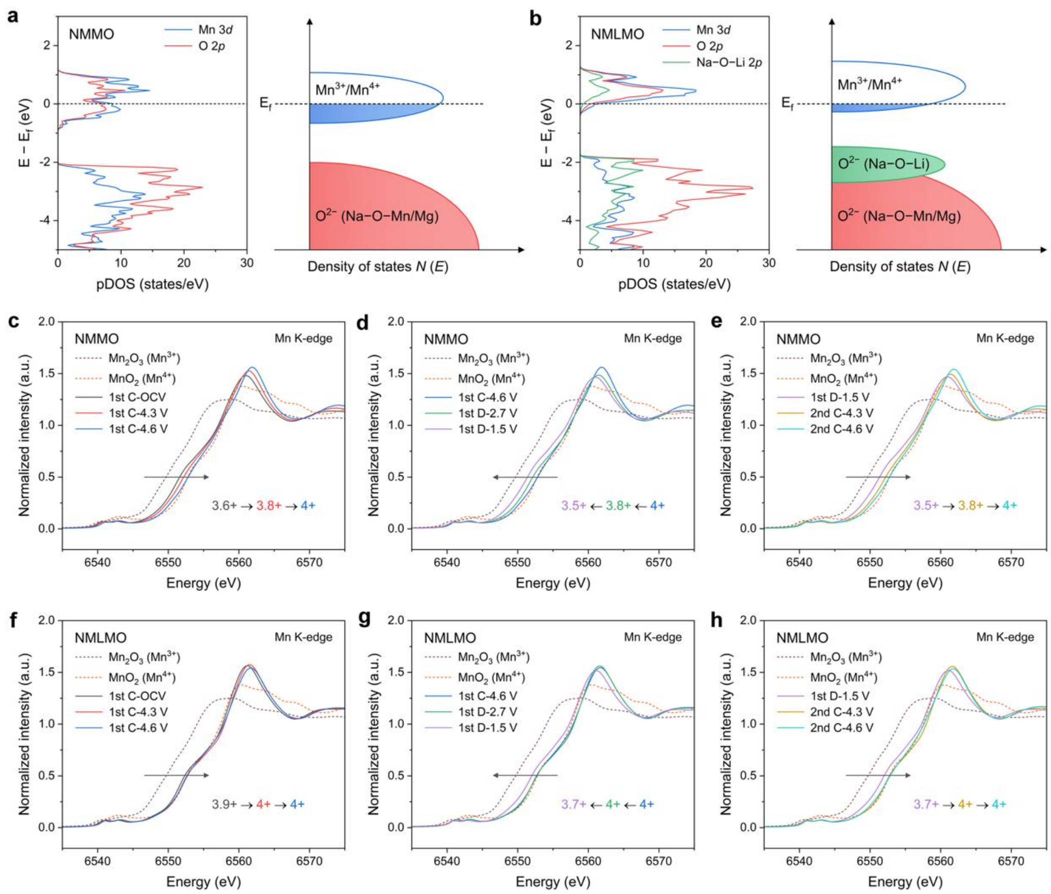

Figure 4. Electronic structure and charge compensation mechanism of NMMO and NMLMO. Calculated pDOS and schematic illustration of the energy versus DOS with the corresponding voltage for (a) NMMO and (b) NMLMO. The Fermi energy is set to 0 eV. Normalized Mn K-edge XANES spectra of (c–e) NMMO and (f–h) NMLMO at different charged and discharged states.

Zero-Strain Characteristic. As the structural stability of cathode materials directly correlates with the electrochemical performance, in-situ XRD technique was conducted to monitor the structural evolution of NMMO and NMLMO in the voltage range of 1.5–4.6 V at 0.05 C (Figures 5a,c, S20 and S21). For NMMO, the position and intensity of (002) and (004) peaks change dramatically during the charge and discharge processes. An additional $(004)_{OP4}$ peak appears at around 16~17° in the high-voltage range of 4.0–4.6 V, signifying the occurrence of P2-OP4 phase transition due to the gliding of TMO_2 slabs. Moreover, in the low-voltage range of 1.5–2.0 V, (104) and (106) peaks gradually disappear with the formation of new (114) and (116) peaks that can be indexed to an orthorhombic P’2 phase (SG: *Cmcm*) derived from the Jahn-Teller distortion of Mn^{3+}. The presence of P2-P’2 phase transition is also evidenced by the splitting of (002) and (004) peaks, as shown in the ex-situ XRD patterns (Figure S22). In contrast, the position and intensity of (002) and (004) peaks for NMLMO exhibit minimal change without any signal of new peaks, symbolizing the successful suppression of high-voltage P2-OP4 and low-voltage P2-P’2 phase transitions. The variation of such characteristic peaks during the second charge mirrors that of the first charge for both cathodes. Obviously, NMLMO experiences a single-phase reaction in a wide voltage range of 1.5–4.6 V (Figure S23), which is different from the biphasic reactions in NMMO and the typical P2-type Mn-based cathodes (*e.g.*, $Na_{2/3}MnO_2$, $Na_{2/3}Ni_{1/3}Mn_{2/3}O_2$ and $Na_xFe_{1/2}Mn_{1/2}O_2$).[48–50]

The lattice parameter variation of NMMO and NMLMO was examined to reflect the structural strain upon charging and discharging (Figure 5b,d). During Na^+ extraction, the c-lattice parameter increases due to the enlarged O^{2-}–O^{2-} electrostatic repulsion between the adjacent oxygen layers, while the a-lattice parameter decreases because of the formation of the electrochemically oxidized TM ions with shrunk ionic radius from Mn^{3+} to Mn^{4+}. Upon Na^+ insertion, both the c- and a-lattice parameters undergo an opposite manner. For NMMO, the c- and a-lattice parameters exhibit a sharp drop in the high-voltage range of 4.0–4.6 V and low-voltage range of 1.5–2.0 V, which is on account of the P2-OP4 and P2-P’2 phase transitions, respectively. These detrimental phase transitions accordingly result in a significant decline of the lattice volume, as highlighted in the biphasic regions (Figure 5b). Comparatively, NMLMO shows smooth lattice parameter variation, confirming the complete single-phase solid-solution reaction. In addition, the maximum lattice volume variation of NMLMO during Na^+ (de)intercalation is calculated to be only 1.2%, which demonstrates the nearly zero-strain characteristic.

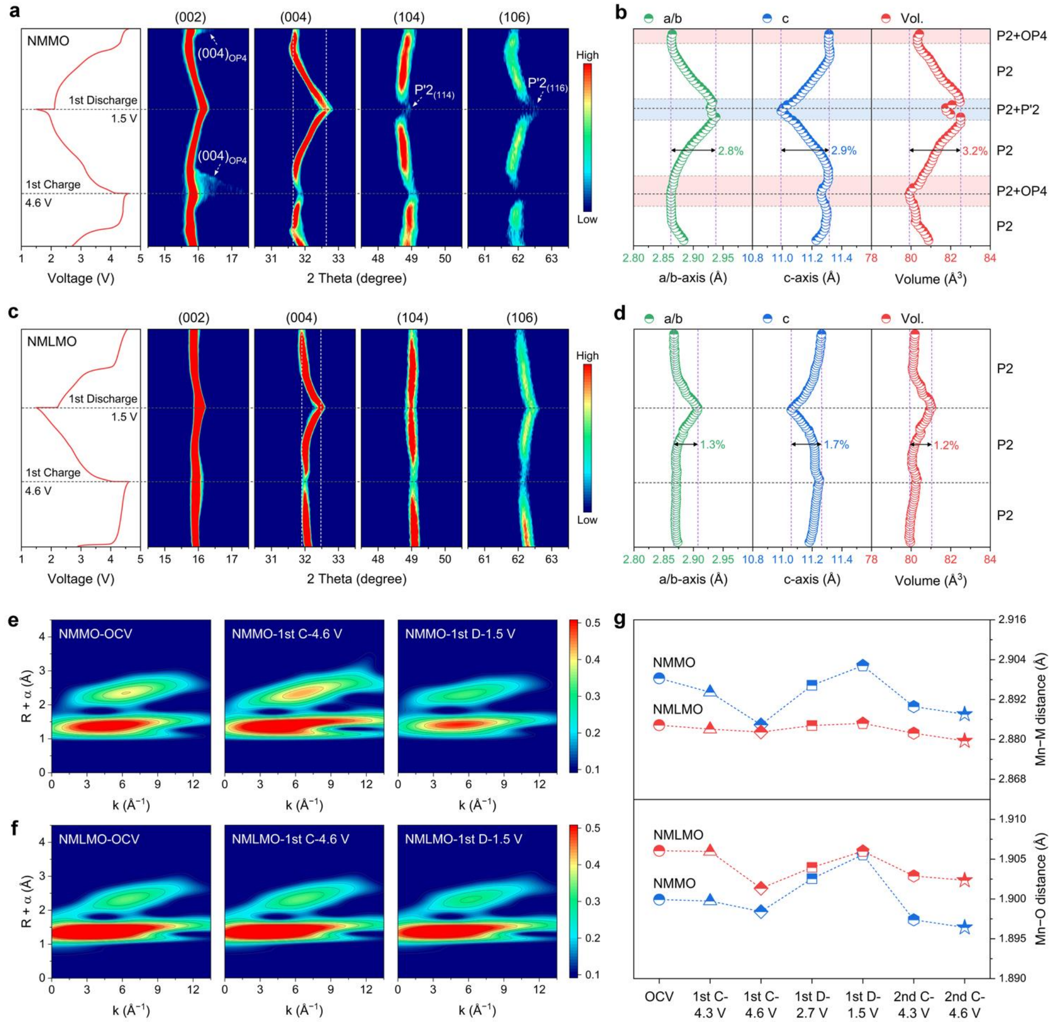

Figure 5. Crystal structure and atomic configuration evolution of NMMO and NMLMO during the charge and discharge processes. Voltage profiles and contour plots of in-situ XRD patterns for (a) NMMO and (c) NMLMO. Lattice parameter change of (b) NMMO and (d) NMLMO. Light red regions indicate the presence of an additional OP4 phase, while light blue region signifies the formation of a P'2 phase. Contour plots of the Mn K-edge WT-EXAFS spectra for (e) NMMO and (f) NMLMO at different charged and discharged states. (g) Interatomic distance of Mn–M (M = Mn, Mg and Li) and Mn–O.

To further reveal the local structural change of NMMO and NMLMO during the charge and discharge processes, the EXAFS characterization was carried out. Figures 5e,f and S24 display the contour plots of Mn K-edge wavelet transformed (WT) EXAFS spectra. Two scattering peaks located at around (5.0 Å^{-1}, 1.5 Å) and (6.5 Å^{-1}, 2.5 Å) can be assigned to the features of Mn–O and Mn–M (M = Mn, Li and Mg) coordination shells, respectively. For NMMO, the shape and intensity of Mn–O and Mn–M features change dramatically upon charging and discharging, which is attributed to the structural disorders resulted from the phase transitions. In contrast, NMLMO exhibits negligible change of the Mn–O and Mn–M features, in good agreement with the FT-EXAFS results (Figure S25). Figure 5g shows the interatomic distance variation of NMMO and NMLMO derived from the Mn K-edge FT-EXAFS fitting data (Figure S26 and Tables S9 and S10). At the OCV state, the average Mn–O and Mn–M distances of NMLMO vary from those of NMMO owing to the different oxidation states of Mn and different TM ordering arrangements in the TM layers. Upon Na^+ deintercalation and intercalation, the Mn–O distance of both cathodes first contracts with the oxidation of Mn and O and then expands with the reduction reaction. Remarkably, the Mn–M distance of NMMO changes significantly due to the occurrence of phase transitions. While, NMLMO presents only a slight change of the Mn–M distance, confirming its high local structural stability.

Combined with the in-situ XRD and EXAFS results, it is concluded that Li ions in the AM sites of NMLMO act as the prismatic pillars to rivet the adjacent TMO_2 slabs and mitigate the Jahn-Teller effect of Mn^{3+}, thus inhibiting the

P2-OP4 and P2-P'2 phase transitions. Furthermore, the morphological evolution of NMMO and NMLMO cathodes during cycling was analyzed by focused ion beam (FIB) SEM to unravel the effect of Li_{AM} (Figure S27). Induced by the progressive phase transitions, severe local distortion and chemo-mechanical cracks appear within the NMMO particles after 50 cycles. By comparison, the single crystals of the cycled NMLMO keep almost intact, which further verifies its nearly zero-strain feature. Altogether, because of the Li_{AM} pillars, NMLMO shows negligible long-range and local structural variation, as well as an absolute single-phase solid-solution reaction during electrochemical cycling in a wide voltage range of 1.5–4.6 V.

DISCUSSION

Here, we demonstrate a Li dual-site substituted P2-$Na_{0.7}Li_{0.03}[Mg_{0.15}Li_{0.07}Mn_{0.75}]O_2$ cathode material with the high-capacity and zero-strain characteristic. The crucial roles of Li_{AM} and Li_{TM} in the charge compensation and the structural stability have been unambiguously disclosed. As depicted in Figure 6a, Li ions in the TM layers construct Na–O–Li configurations to increase the capacity from oxygen redox, which are similar to the typical Li–O–Li configurations of Li-rich cathodes in Li-ion batteries.[51–53] Meanwhile, Li ions in the AM layers serve as LiO_6 prismatic pillars to restrain the phase transitions and mitigate the lattice strain. NMLMO, thereby, exhibits a high specific capacity (266 mAh g^{-1}) with the minimal lattice volume variation (1.2%) during the dynamic (de)sodiation processes.

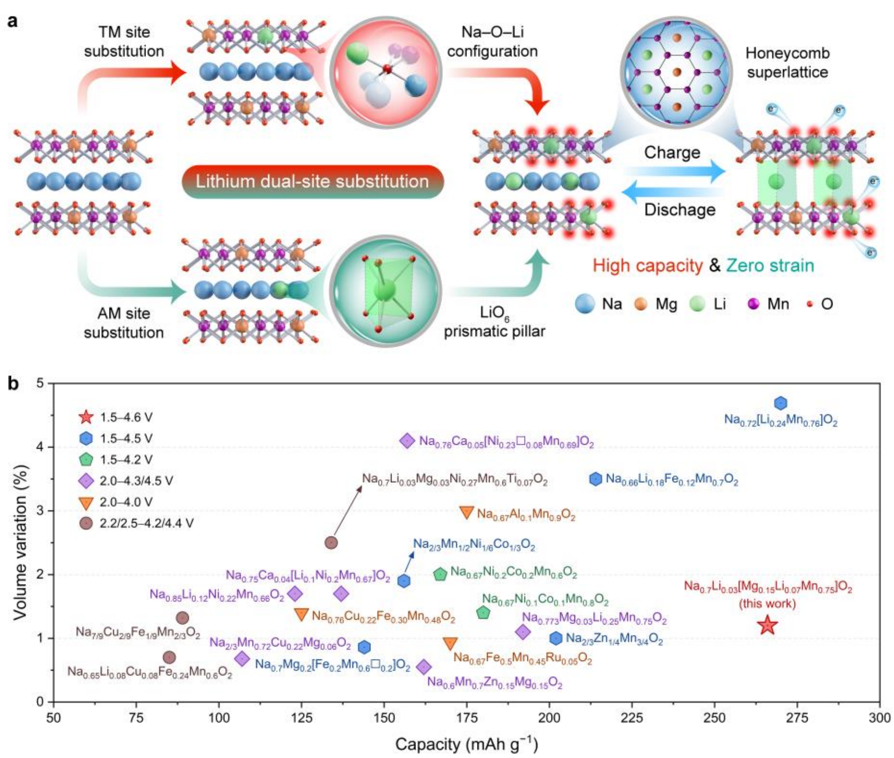

Figure 6. Effects of Li_{TM} and Li_{AM} on charge compensation and structural stability for NMLMO during the charge and discharge processes. (a) Schematic diagram for the mechanism of Li_{TM} and Li_{AM} to realize the high-capacity and zero-strain characteristic of NMLMO. (b) Comparison of capacity versus volume variation in various P2-type oxide cathode materials.

The local Li environment in P2-type hosts is a key factor that affects the structural stability.[54–56] In absence of the Li loss in the P2 structure as detected by ICP-OES (Table S11), we further performed the ^{7}Li MAS NMR measurement to understand the fate of Li in NLMMO during the charge and discharge processes. Unlike those previously reported Li-substituted Na-layered oxide cathodes in which Li dynamically migrates between the TM and AM layers, the content ratio of Li_{TM} and Li_{AM} in NMLMO shows only a slight change, indicating the high stability of Li (Figure S28).[18,29] This is probably because the complete single-phase solid-solution behavior eliminates the interlayer migration pathways (*e.g.*, tetrahedral AM sites in the O2/OP4 intermediate phase). Operando differential electrochemical mass spectrometry (DEMS) was carried out to evaluate the stability of lattice oxygen (Figure S29). No detectable O_2 gas is generated from NMLMO in the first cycle, which indicates the absence of lattice oxygen release. Though the oxygen redox activity in NMLMO is greatly enhanced by Li substitution, the oxygen redox reversibility is still guaranteed by its stable anionic sublattice. Moreover, the honeycomb superstructure of NMLMO is also retained upon charging and discharging, as confirmed by ex-situ XRD (Figures S30 and S31). These above results manifest that the Li ions almost remain in their original TM and AM sites of the P2 structure even in a wide voltage of 1.5–4.6 V.

The comparison of capacity and volume variation between NMLMO and the previously reported P2-type oxide cathode materials during the charge and discharge processes is summarized in Figure 6b and Table S8. Note that the Na content also plays a critical role in the structural stability of P2 hosts, as Na ions can shield the electrostatic repulsions between the oxygen layers. Most of the P2-type oxide cathodes show low lattice strain in a confined voltage range under 4.2 V, which is due to the mitigation of the gliding of TMO_2 slabs by the retained Na ions in the AM layers. Though the unfavorable phase transitions are delayed, these cathodes deliver capacities of less than 200 mAh g^{-1}. The reported $Na_{0.72}Li_{0.24}Mn_{0.76}O_2$ achieves a relatively high capacity (270 mAh g^{-1}) in a wide voltage of 1.5–4.5 V, but it suffers from large lattice volume change (4.69%) and fast capacity decay.[19] Comparatively, the Li dual-site substituted NMLMO not only delivers a high capacity of 266 mAh g^{-1} based on both cationic (Mn) and anionic (O) redox, but also features the quasi-zero-strain characteristic with only 1.2% lattice volume variation. The entire P2 phase of NMLMO can be well maintained upon the large amount of Na^+ (~0.93 Na^+ per formula unit) extraction and insertion. Overall, through the synergistic effect of Li_{TM} and Li_{AM}, NMLMO simultaneously realizes high energy density and high electrochemical stability.

For potential practical implementations, the air/water stability of NMLMO was further investigated (Figure S32). The P2 phase is well preserved for both air-exposed and water-treated NMLMO, which demonstrates its superior structural stability in ambient air and even after water treatment. The excellent air/water stability of NMLMO originates from the enhanced oxygen redox reactivity and the reduced interlayer spacing by the Li dual-site substitution.[57]

CONCLUSION

In summary, to counter the capacity-stability trade-off dilemma, we have developed the Li dual-site substituted P2-$Na_{0.7}Li_{0.03}[Mg_{0.15}Li_{0.07}Mn_{0.75}]O_2$ as a high-capacity zero-strain cathode material for SIBs. The well-deigned NMLMO possesses a high specific capacity of 266 mAh g^{-1} and exhibits excellent structural stability in a wide voltage range of 1.5–4.6 V. Combined DFT calculations and XANES measurements demonstrate that Li ions in the octahedral TM sites trigger additional unhybridized O 2*p* orbitals to promote the capacity contribution of oxygen redox. Meanwhile, as evidenced by in-situ XRD and EXAFS characterizations, Li ions in the trigonal prismatic AM sites function as rigid pillars to suppress the P2-OP4 and P2-P'2 phase transitions, which renders the long-range and local structure highly stable with only 1.2% lattice volume variation during the charge and discharge processes. This research offers new insights into simultaneously increasing the specific capacity and minimizing the structural strain for advanced cathode materials.

ASSOCIATED CONTENT

Supporting Information

Details of preparations, characterizations, measurements, and calculations; additional NMR, XPS, XAS, SEM, TEM, XRD, NPD, EDS, DEMS, supplementary electrochemical results and tables. The Supporting Information is available free of charge at http://pubs.acs.org.

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Notes

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REFERENCES

(1) Usiskin, R.; Lu, Y.; Popovic, J.; Law, M.; Balaya, P.; Hu, Y.-S.; Maier, J. Fundamentals, status and promise of sodium-based batteries. *Nat. Rev. Mater.* **2021**, *6*, 1020–1035.

(2) Hwang, J.-Y.; Myung, S.-T.; Sun, Y.-K. Sodium-ion batteries: present and future. *Chem. Soc. Rev.* **2017**, *46*, 3529–3614.

(3) Vaalma, C.; Buchholz, D.; Weil, M.; Passerini, S. A cost and resource analysis of sodium-ion batteries. *Nat. Rev. Mater.* **2018**, 3, 18013.

(4) Tian, Y.; Zeng, G.; Rutt, A.; Shi, T.; Kim, H.; Wang, J.; Koettgen, J.; Sun, Y.; Ouyang, B.; Chen, T.; Lun, Z.; Rong, Z.; Persson, K.; Ceder, G. Promises and Challenges of Next-Generation "Beyond Li-ion" Batteries for Electric Vehicles and Grid Decarbonization. *Chem. Rev.* **2021**, *121*, 1623-1669.

(5) Deng, J.; Luo, W.-B.; Chou, S.-L.; Liu, H.-K.; Dou, S.-X. Sodium-Ion Batteries: From Academic Research to Practical Commercialization. *Adv. Energy Mater.* **2018**, *8*, 1701428.

(6) Xiang, X.; Zhang, K.; Chen, J. Recent Advances and Prospects of Cathode Materials for Sodium-Ion Batteries. *Adv. Mater.* **2015**, *27*, 5343–5364.

(7) Li, Y.; Lu, Y.; Zhao, C.; Hu, Y.-S.; Titirici, M.-M.; Li, H.; Huang, X.; Chen, L. Recent advances of electrode materials for low-cost sodium-ion batteries towards practical application for grid energy storage. *Energy Storage Mater.* **2017**, *7*, 130–151.

(8) Chen, M.; Liu, Q.; Wang, S.-W.; Wang, E.; Guo, X.; Chou, S.-L. High-Abundance and Low-Cost Metal-Based Cathode Materials for Sodium-Ion Batteries: Problems, Progress, and Key Technologies. *Adv. Energy Mater.* **2019**, *9*, 1803609.

(9) Zhao, C.; Wang, Q.; Yao, Z.; Wang, J.; Sánchez-Lengeling, B.; Ding, F.; Qi, X.; Lu, Y.; Bai, X.; Li, B.; Li, H.; Aspuru-Guzik, A.; Huang, X.; Delmas, C.; Wagemaker, M.; Chen, L.; Hu, Y.-

S. Rational design of layered oxide materials for sodium-ion batteries. *Science* **2020**, *370*, 708–711.
(10) Wang, P.-F.; You, Y.; Yin, Y.-X.; Guo, Y.-G. Layered Oxide Cathodes for Sodium-Ion Batteries: Phase Transition, Air Stability, and Performance. *Adv. Energy Mater.* **2018**, *8*, 1701912.
(11) Jin, T.; Li, H.; Zhu, K.; Wang, P.-F.; Liu, P.; Jiao, L. Polyanion-type cathode materials for sodium-ion batteries. *Chem. Soc. Rev.* **2020**, *49*, 2342–2377.
(12) Liu, Y.; Li, J.; Shen, Q.; Zhang, J.; He, P.; Q, X.; Liu, Y. Advanced characterizations and measurements for sodium-ion batteries with NASICON-type cathode materials. *eScience* **2022**, *2*, 10–30.
(13) Qian, J.; Wu, C.; Cao, Y.; Ma, Z.; Huang, Y.; Ai, X.; Yang, H. Prussian Blue Cathode Materials for Sodium-Ion Batteries and Other Ion Batteries. *Adv. Energy Mater.* **2018**, *8*, 1702619.
(14) Wang, W.; Gang, Y.; Hu, Z.; Yan, Z.; Li, W.; Li, Y.; Gu, Q.-F.; Wang, Z.; Chou, S.-L.; Liu, H.-K.; Dou, S.-X. Reversible structural evolution of sodium-rich rhombohedral Prussian blue for sodium-ion batteries. *Nat. Commun.* **2020**, *11*, 980.
(15) Zhao, Q.; Lu, Y.; Chen, J. Advanced Organic Electrode Materials for Rechargeable Sodium-Ion Batteries. *Adv. Energy Mater.* **2017**, *7*, 1601792.
(16) Wang, X.; Shang, Z.; Yang, A.; Zhang, Q.; Cheng, F.; Jia, D.; Chen, J. Combining Quinone Cathode and Ionic Liquid Electrolyte for Organic Sodium-Ion Batteries. *Chem* **2019**, *5*, 364–375.
(17) Liu, Q.; Hu, Z.; Li, W.; Zou, C.; Jin, H.; Wang, S.; Chou, S.; Dou, S.-X. Sodium transition metal oxides: the preferred cathode choice for future sodium-ion batteries? *Energy Environ. Sci.* **2021**, *14*, 158–179.
(18) Wang, Q.; Mariyappan, S.; Rousse, G.; Morozov, A. V.; Porcheron, B., Dedryvère, R., Wu, J., Yang, W., Zhang, L., Chakir, M.; Avdeev, M.; Deschamps, M.; Yu, Y.-S.; Cabana, J.; Doublet, M.-L.; Abakumov, A. M.; Tarascon, J.-M. Unlocking anionic redox activity in O3-type sodium 3d layered oxides via Li substitution. *Nat. Mater.* **2021**, *20*, 353–361.
(19) Rong, X.; Hu, E.; Lu, Y.; Meng, F.; Zhao, C.; Wang, X.; Zhang, Q.; Yu, X.; Gu, L.; Hu, Y.-S.; Li, H.; Huang, X.; Yang, X.-Q.; Delmas, C.; Chen, L. Anionic Redox Reaction-Induced High-Capacity and Low-Strain Cathode with Suppressed Phase Transition. *Joule* **2019**, *3*, 503–517.
(20) Maitra, U.; House, R. A.; Somerville, J. W.; Tapia-Ruiz, N.; Lozano, J. G.; Guerrini, N.; Hao, R.; Luo, K.; Jin, L.; Perez-Osorio, M. A.; Massel, F.; Pickup, D. M.; Ramos, S.; Lu, X.; McNally, D. E.; Chadwick, A. V.; Giustino, F.; Schmitt, T.; Duda, L. C.; Roberts, M. R.; Bruce, P. G. Oxygen redox chemistry without excess alkali-metal ions in $Na_{2/3}[Mg_{0.28}Mn_{0.72}]O_2$. *Nat. Chem.* **2018**, *10*, 288–295.
(21) Rahman, M. M.; Lin, F. Oxygen Redox Chemistry in Rechargeable Li-Ion and Na-Ion Batteries. *Matter* **2021**, *4*, 490–527.
(22) Vergnet, J.; Saubanère, M.; Doublet, M.-L.; Tarascon, J.-M. The Structural Stability of P2-Layered Na-Based Electrodes during Anionic Redox. *Joule* **2020**, *4*, 420–434.
(23) Sun, Y.; Guo, S.; Zhou, H. Adverse effects of interlayer-gliding in layered transition-metal oxides on electrochemical sodium-ion storage. *Energy Environ. Sci.* **2019**, *12*, 825–840.
(24) Somerville, J. W.; Sobkowiak, A.; Tapia-Ruiz, N.; Billaud, J.; Lozano, J. G.; House, R. A.; Gallington, L. C.; Ericsson, T.; Häggström, L.; Roberts, M. R.; Maitra, U.; Bruce, P. G. Nature of the "Z"-phase in layered Na-ion battery cathodes. *Energy Environ. Sci.* **2019**, *12*, 2223–2232.
(25) Li, X.; Ma, X.; Su, D.; Liu, L.; Chisnell, R.; Ong, S. P.; Chen, H.; Toumar, A.; Idrobo, J.-C.; Lei, Y.; Bai, J.; Wang, F.; Lynn, J. W.; Lee, Y. S.; Ceder, G. Direct visualization of the Jahn-Teller effect coupled to Na ordering in $Na_{5/8}MnO_2$. *Nat. Mater.* **2014**, *13*, 586–592.
(26) Bai, X.; Iadecola, A.; Tarascon, J.-M.; Rozier, P. Decoupling the effect of vacancies and electropositive cations on the anionic redox processes in Na based P2-type layered oxides. *Energy Storage Mater.* **2020**, *31*, 146–155.
(27) Kim, B.; Song, J.-H.; Eum, D.; Yu, S.; Oh, K.; Lee, M. H.; Jang, H.-Y.; Kang, K. A theoretical framework for oxygen redox chemistry for sustainable batteries. *Nat. Sustain.* **2022**, *5*, 708–716.
(28) Gao, A.; Zhang, Q.; Li, X.; Shang, T.; Tang, Z.; Lu, X.; Luo, Y.; Ding, J.; Kan, W. H.; Chen, H.; Yin, W.; Wang, X.; Xiao, D.; Su, D.; Li, H.; Rong, X.; Yu, X.; Yu, Q.; Meng, F.; Nan, C.; Delmas, C.; Chen, L; Hu, Y.-S.; Gu, Lin. Topologically protected oxygen redox in a layered manganese oxide cathode for sustainable batteries. *Nat. Sustain.* **2022**, *5*, 214–224.
(29) House, R. A.; Maitra, U.; Perez-Osorio, M. A.; Lozano, J. G.; Jin, L.; Somerville, J. W.; Duda, L. C.; Nag, A.; Walters, A.; Zhou, K. J.; Roberts, M. R.; Bruce, P. G. Superstructure control of first-cycle voltage hysteresis in oxygen-redox cathodes. *Nature* **2020**, *577*, 502–508.
(30) McColl, K.; House, R. A.; Rees, G. J.; Squires, A. G.; Coles, S. W.; Bruce, P. G.; Morgan, B. J.; Islam, M. S. Transition metal migration and O_2 formation underpin voltage hysteresis in oxygen-redox disordered rocksalt cathodes. *Nat. Commun.* **2022**, *13*, 5275.
(31) Wang, K.; Zhang, Z.; Cheng, S.; Han, X.; Fu, J.; Sui, M.; Yan, P. Precipitate-stabilized surface enabling high-performance $Na_{0.67}Ni_{0.33-x}Mn_{0.67}Zn_xO_2$ for sodium-ion battery. *eScience* **2022**, *2*, 529–536.
(32) Ren, M. Zhao, S., Gao, S., Zhang, T., Hou, M., Zhang, W., Feng, K.; Zhong, J. Hua, W.; Indris S.; Zhang, K.; Chen, J.; Li, F. Homeostatic Solid Solution in Layered Transition-Metal Oxide Cathodes of Sodium-Ion Batteries. *J. Am. Chem. Soc.* **2023**, *145*, 224–233.
(33) Zhao, C.; Yao, Z.; Wang, Q.; Li, H.; Wang, J.; Liu, M.; Ganapathy, S.; Lu, Y.; Cabana, J.; Li, B.; Bai, X.; Aspuru-Guzik, A.; Wagemaker, M.; Chen, L.; Hu, Y.-S. Revealing High Na-Content P2-Type Layered Oxides as Advanced Sodium-Ion Cathodes. *J. Am. Chem. Soc.* **2020**, *142*, 5742–5750.
(34) Wang, Y.; Yu, X.; Xu, S.; Bai, J.; Xiao, R.; Hu, Y.-S.; Li, H.; Yang, X.-Q.; Chen, L.; Huang, X. A zero-strain layered metal oxide as the negative electrode for long-life sodium-ion batteries. *Nat. Commun.* **2013**, *4*, 2365.
(35) Wang, C.; Liu, L.; Zhao, S.; Liu, Y.; Yang, Y.; Yu, H.; Lee, S.; Lee, G.-H.; Kang, Y.-M.; Liu, R.; Li, F.; Chen, J. Tuning local chemistry of P2 layered-oxide cathode for high energy and long cycles of sodium-ion battery. *Nat. Commun.* **2021**, *12*, 2256.
(36) Wang, Q.-C.; Meng, J.-K.; Yue, X.-Y.; Qiu, Q.-Q.; Song, Y.; Wu, X.-J.; Fu, Z.-W.; Xia, Y.-Y.; Shadike, Z.; Wu, J.; Yang, X.-Q.; Zhou, Y.-N. Tuning P2-Structured Cathode Material by Na-Site Mg Substitution for Na-Ion Batteries. *J. Am. Chem. Soc.* **2019**, *141*, 840–848.
(37) Zhang, K.; Kim, D.; Hu, Z.; Park, M.; Noh, G.; Yang, Y.; Zhang, J.; Lau, V. W.; Chou, S.-L.; Cho, M.; Choi, S.-Y.; Kang, Y.-M. Manganese based layered oxides with modulated electronic and thermodynamic properties for sodium ion batteries. *Nat. Commun.* **2019**, *10*, 5203.
(38) Wang, Y.; Zhao, X.; Jin, J.; Shen, Q.; Zhang, N.; Qu, X.; Liu, Y.; Jiao, L. Low-cost layered oxide cathode involving cationic and anionic redox with a complete solid-solution sodium-storage behavior. *Energy Storage Mater.* **2022**, *47*, 44–50.

(39) Chu, S.; Zhang, C.; Xu, H.; Guo, S.; Wang, P.; Zhou, H. Pinning Effect Enhanced Structural Stability toward a Zero-Strain Layered Cathode for Sodium-Ion Batteries. *Angew. Chem. Int. Ed.* **2021**, *60*, 13366–13371.

(40) Yabuuchi, N.; Hara, R.; Kajiyama, M.; Kubota, K.; Ishigaki, T.; Hoshikawa, A.; Komaba, S. New O2/P2-type Li-Excess Layered Manganese Oxides as Promising Multi-Functional Electrode Materials for Rechargeable Li/Na Batteries. *Adv. Energy Mater.* **2014**, *4*, 1301453.

(41) Berthelot, R.; Carlier, D.; Delmas, C. Electrochemical investigation of the P2-Na_xCoO_2 phase diagram. *Nat. Mater.* **2011**, *10*, 74–80.

(42) Yang, L.; Amo, J. M. L.; Shadike, Z.; Bak, S.-M.; Bonilla, F.; Galceran, M.; Nayak, P. K.; Buchheim, J. R.; Yang, X.-Q.; Rojo, T.; Adelhelm, P. A Co- and Ni-Free P2/O3 Biphasic Lithium Stabilized Layered Oxide for Sodium-Ion Batteries and its Cycling Behavior. *Adv. Funct. Mater.* **2020**, *30*, 2003364.

(43) Xiao, B.; Wang, Y.; Tan, S.; Song, M.; Li, X.; Zhang, Y.; Lin, F.; Han, K. S.; Omenya, F.; Amine, K.; Yang, X.-Q.; Reed, D.; Hu, Y.; Xu, G.-L.; Hu, E.; Li, X.; Li, X. Vacancy-Enabled O3 Phase Stabilization for Manganese-Rich Layered Sodium Cathodes. *Angew. Chem. Int. Ed.* **2021**, *60*, 8258–8267.

(44) Yang, L.; Li, X.; Liu, J.; Xiong, S.; Ma, X.; Liu, P.; Bai, J.; Xu, W.; Tang, Y.; Hu, Y.-Y.; Liu, M.; Chen, H. Lithium-Doping Stabilized High-Performance P2-$Na_{0.66}Li_{0.18}Fe_{0.12}Mn_{0.7}O_2$ Cathode for Sodium Ion Batteries. *J. Am. Chem. Soc.* **2019**, *141*, 6680–6689.

(45) Li, C.; Geng, F.; Hu, B.; Hu, B. Anionic redox in Na-based layered oxide cathodes: a review focus on mechanism studies. Mater. Today Energy **2020**, *17*, 100474.

(46) Yang, L; Chen, R.; Liu, Z.; Gao, Y.; Wang X.; Wang, Z.; Chen, L. Configuration-dependent anionic redox in cathode materials. Battery Energy **2022**, *1*, 20210015.

(47) Cao, X.; Li, X.; Qiao, Y.; Jia, M.; Qiu, F.; He, Y.; He, P.; Zhou, H. Restraining Oxygen Loss and Suppressing Structural Distortion in a Newly Ti-Substituted Layered Oxide P2-$Na_{0.66}Li_{0.22}Ti_{0.15}Mn_{0.63}O_2$. ACS Energy Lett. **2019**, *4*, 2409–2417.

(48) Kumakura, S.; Tahara, Y.; Kubota, K.; Chihara, K.; Komaba, S. Sodium and Manganese Stoichiometry of P2-Type $Na_{2/3}MnO_2$. *Angew. Chem. Int. Ed.* **2016**, 55, 12760–12763.

(49) Lu, Z.; Dahn, J. R. *In Situ* X-Ray Diffraction Study of P2-$Na_{2/3}[Ni_{1/3}Mn_{2/3}]O_2$. *J. Electrochem. Soc.* **2001**, *148*, A1225–A1229.

(50) Yabuuchi, N.; Kajiyama, M.; Iwatate, J.; Nishikawa, H.; Hitomi, S.; Okuyama, R.; Usui, R.; Yamada, Y.; Komaba, S. P2-type $Na_x[Fe_{1/2}Mn_{1/2}]O_2$ made from earth-abundant elements for rechargeable Na batteries. *Nat. Mater.* **2012**, *11*, 512–517.

(51) Seo, D.-H.; Lee, J.; Urban, A.; Malik, R.; Kang, S.; Ceder, G. The structural and chemical origin of the oxygen redox activity in layered and cation-disordered Li-excess cathode materials. *Nat. Chem.* **2016**, *8*, 692–697.

(52) Assat, G.; Tarascon, J.-M. Fundamental understanding and practical challenges of anionic redox activity in Li-ion batteries. *Nat. Energy* **2018**, *3*, 373–386.

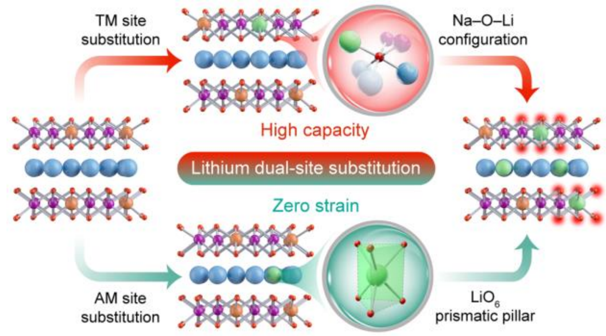
(53) Wang, M.; Chen, X.; Yao, H.; Lin, G.; Lee, J.; Chen, Y.; Chen, Q. Research Progress in Lithium-Excess Disordered Rock-Salt Oxides Cathode. *Energy Environ. Mater.* **2022**, 5, 1139-1154.

(54) Voronina, N.; Shin, M.-Y.; Kim, H.-J.; Yaqoob, N.; Guillon, O. Song, S. H.; Kim, H.; Lim, H.-D.; Jung, H.-G.; Kim, Y.; Lee, H.-K.; Lee, K.-S.; Yazawa, K.; Gotoh, K.; Kaghazchi, P.; Myung, S.-T. Hysteresis-Suppressed Reversible Oxygen-Redox Cathodes for Sodium-Ion Batteries. *Adv. Energy Mater.* **2022**, *12*, 2103939.

(55) Qi, X.; Wu, L.; Li, Z.; Xiang, Y.; Liu, Y.; Huang, K.; Yuval, E.; Aurbach, D.; Zhang, X. Superstructure Variation and Improved Cycling of Anion Redox Active Sodium Manganese Oxides Due to Doping by Iron. *Adv. Energy Mater.* **2022**, *12*, 2202355.

(56) Zhao, C.; Li, C.; Liu, H.; Qiu, Q.; Geng, F.; Shen, M.; Tong, W.; Li, J.; Hu, B. Coexistence of $(O_2)^{n-}$ and Trapped Molecular O_2 as the Oxidized Species in P2-Type Sodium 3d Layered Oxide and Stable Interface Enabled by Highly Fluorinated Electrolyte. *J. Am. Chem. Soc.* **2021**, *143*, 18652–18664.

(57) Zuo, W.; Qiu, J.; Liu, X.; Ren, F.; Liu, H.; He, H.; Luo, C.; Li, J.; Ortiz, G. F.; Duan, H.; Liu, J.; Wang, M.-S.; Li, Y.; Fu, R.; Yang, Y. The stability of P2-layered sodium transition metal oxides in ambient atmospheres. *Nat. Commun.* **2020** *11*, 3544.


TM site
substitution
Na–O–Li
configuration
High capacity
Lithium dual-site substitution
Zero strain
AM site
substitution
LiO6
prismatic pillar

TOC Graphic